

IT Applications in Business

Services Management in the enterprise

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HR EXCELLENCE IN RESEARCH



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IT Infrastructure

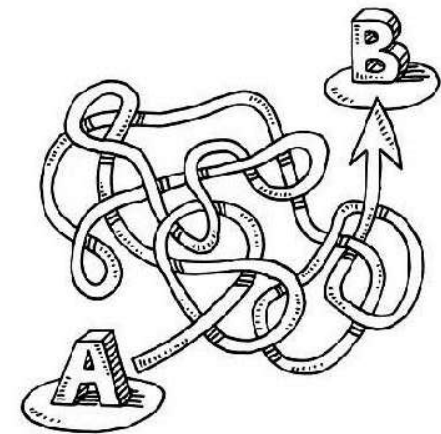
- Hardware
- Software
- Connectivity
- Procedures
- Documentation
- ***People***



What is *ITIL*?

- The IT Infrastructure Library (ITIL) is a library of volumes describing a framework of **best practices*** for delivering IT services

*A method or technique that has consistently shown results superior to those achieved with other means, and that is used as a benchmark.



What is *ITIL*?

- Created based on the best industry practices in IT
- Defines a common language for IT and business
- Platform Independent
- ITIL 4 was released on February 28, 2019
- Developed by the British government's Central Computer and Telecommunications Agency (CCTA) (which merged with Office of Government Commerce) during the 1980s

What is *ITIL*?

- ITIL provides a systematic and professional approach to the management of IT service provision, and offers the following benefits:
 - reduced IT costs
 - improved IT services through the use of proven best practice processes
 - improved customer satisfaction through a more professional approach to service delivery
 - standards and guidance
 - improved productivity
 - improved use of skills and experience
 - improved delivery of third-party services through the specification of ITIL or BS15000 as the standard for service delivery in services procurements

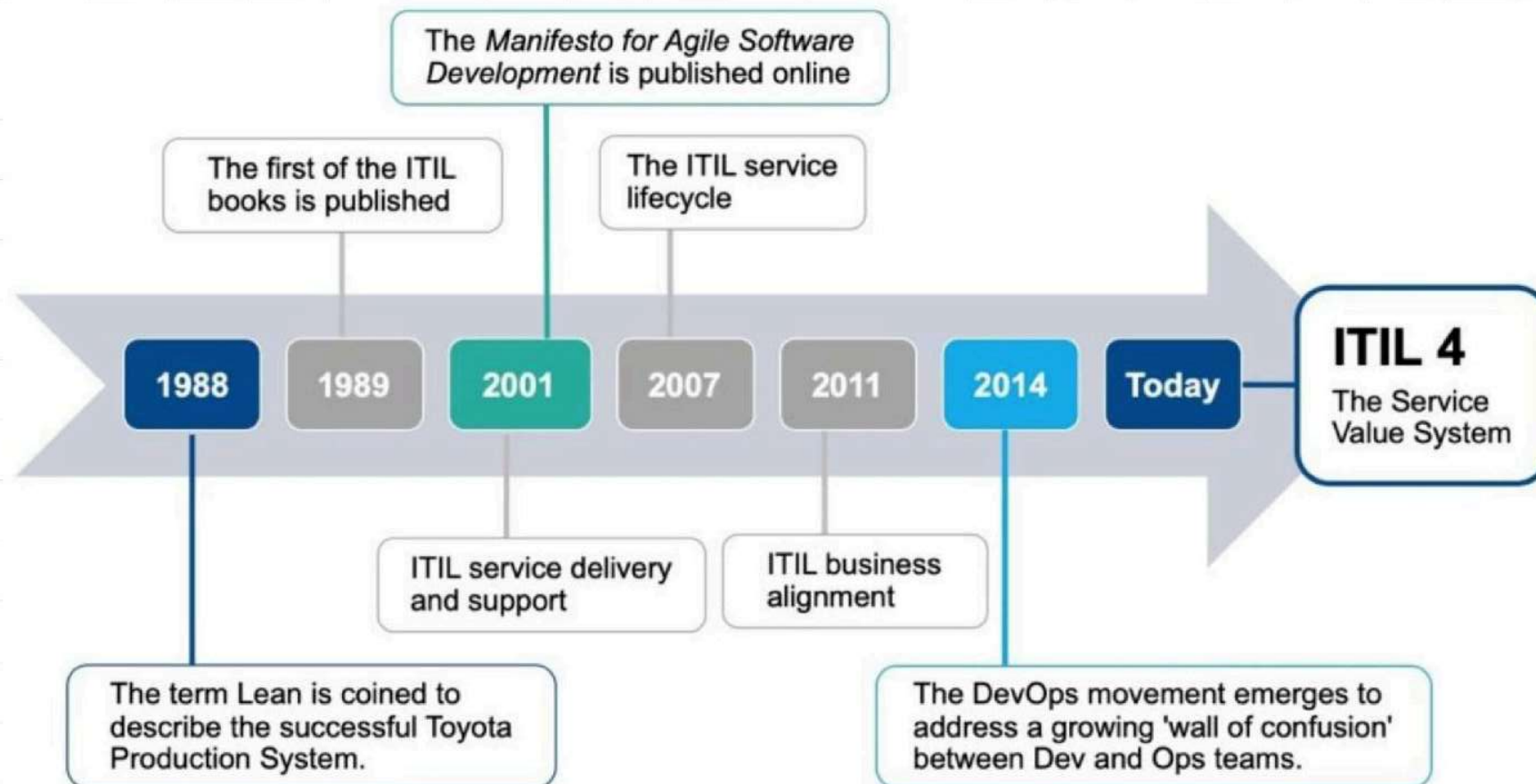
What is ITSM ...



- ***IT Service Management***

- a set of specialized organizational capabilities for providing value to customers in the form of IT services
- implementation and management of quality IT services that meet the needs of the business
- carried out by IT service providers through an appropriate mix of people, process, partners, and information technology

- ITIL is the preeminent best practice framework for ITSM



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ITIL v3

- Process model created on the basis of best industry practices:
 - *Service Delivery*
 - *Service Support*
 - *Application management*
 - *Security management*

What is *ITIL*?

UPs

- ✓ Process description standard
- ✓ Focus on processes, not technologies
- ✓ Integrates processes
- ✓ Standardization makes it possible to improve the quality of services and increase profitability
- ✓ **Customer focus (through *service*)**

DOWNs

- It is not a methodology for streamlining processes
- **Defines "*what*" but does not specify "*how*"**
- It does not include "all" processes
- It does not cover organizational issues
- media noise leads to unrealistic expectations

What is NOT?

- **NOT** a "universal tool" for solving problems
- **NOT** the norm for strict enforcement - a collection of best practices
- Does **NOT** define specific procedures
- Strives to improve efficiency and increase efficiency
- Minimizes risks
- Meets the requirements of **ISO 9001** quality standards



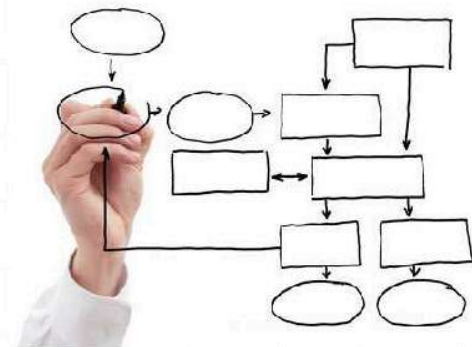
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Service Management in practice

- **Process**

- a set of actions aimed at achieving specific goals and providing value to clients or stakeholders
- A strategic advantage in diversifying the market and creating an advantage over the competition
 - Measurable
 - Output = concrete results
 - It provides results to clients / interested parties
 - Reacts to specific events

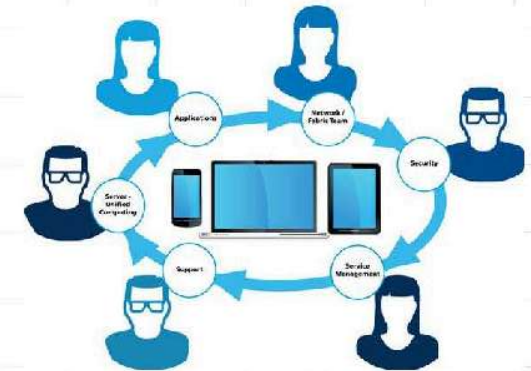


Service Management in practice



- **Service owner, Service manager**
 - responsible for establishing responsibilities for the lifecycle of specific services
- **Process owner, Process manager, Process practitioner**
 - helping to identify responsibilities in the ITSM processes management and execution
- **Process practitioner: (e.g. Configuration manager, Capacity manager, Service Desk specialist)**
 - describing specific responsibilities within certain processes and functions

Service Management in practice



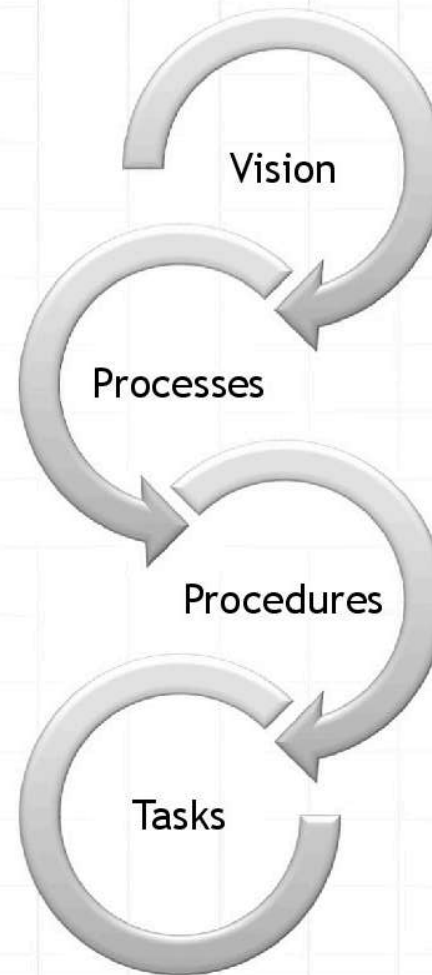
- **Function**

- a team or group of people and tools used to perform one or more processes or activities

- **Role**

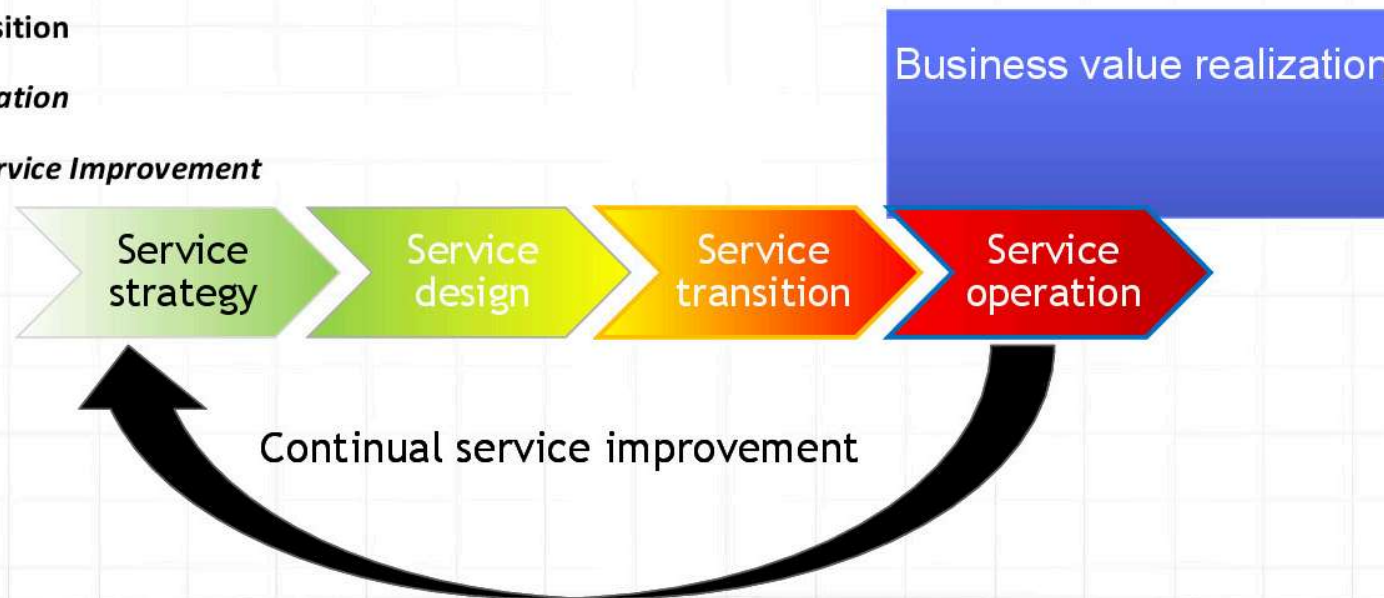
- a set of duties, actions and authorities granted to a person or team

Service model



Service Lifecycle

- 5 stages of the service life cycle = 5 basic ITILv3 books
 - *Service Strategy*
 - *Service Design*
 - *Service Transition*
 - *Service Operation*
 - *Continual Service Improvement*



Service Lifecycle

SERVICE STRATEGY

- Service Strategy for IT Services
- Service Portfolio Management
- Business Relationship Management
- Financial Management for IT Services
- Demand Management

SERVICE OPERATION

- Event Management
- **Incident Management**
- Request Fulfillment
- **Problem Management**
- Access Management

Functions

- Service Desk
- Technical Management
- IT Operations Management
- Application Management

CONTINUAL SERVICE IMPROVEMENT

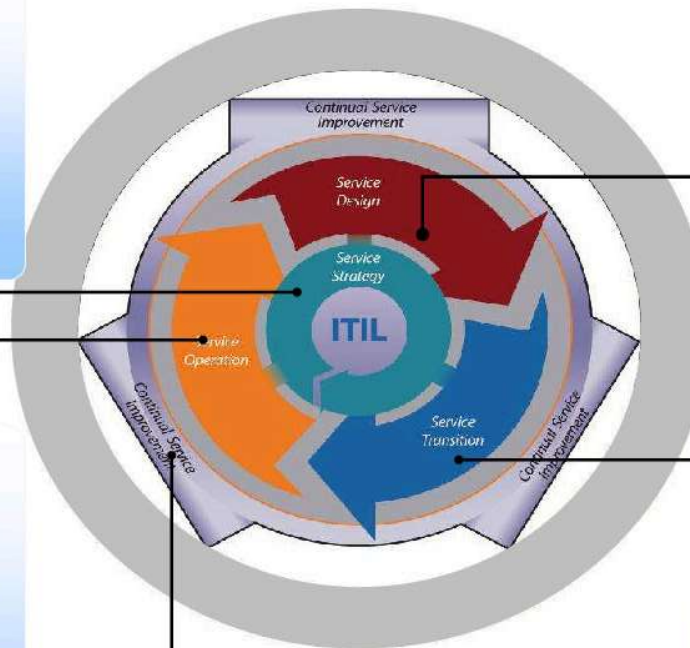
- Seven Step Improvement

SERVICE DESIGN

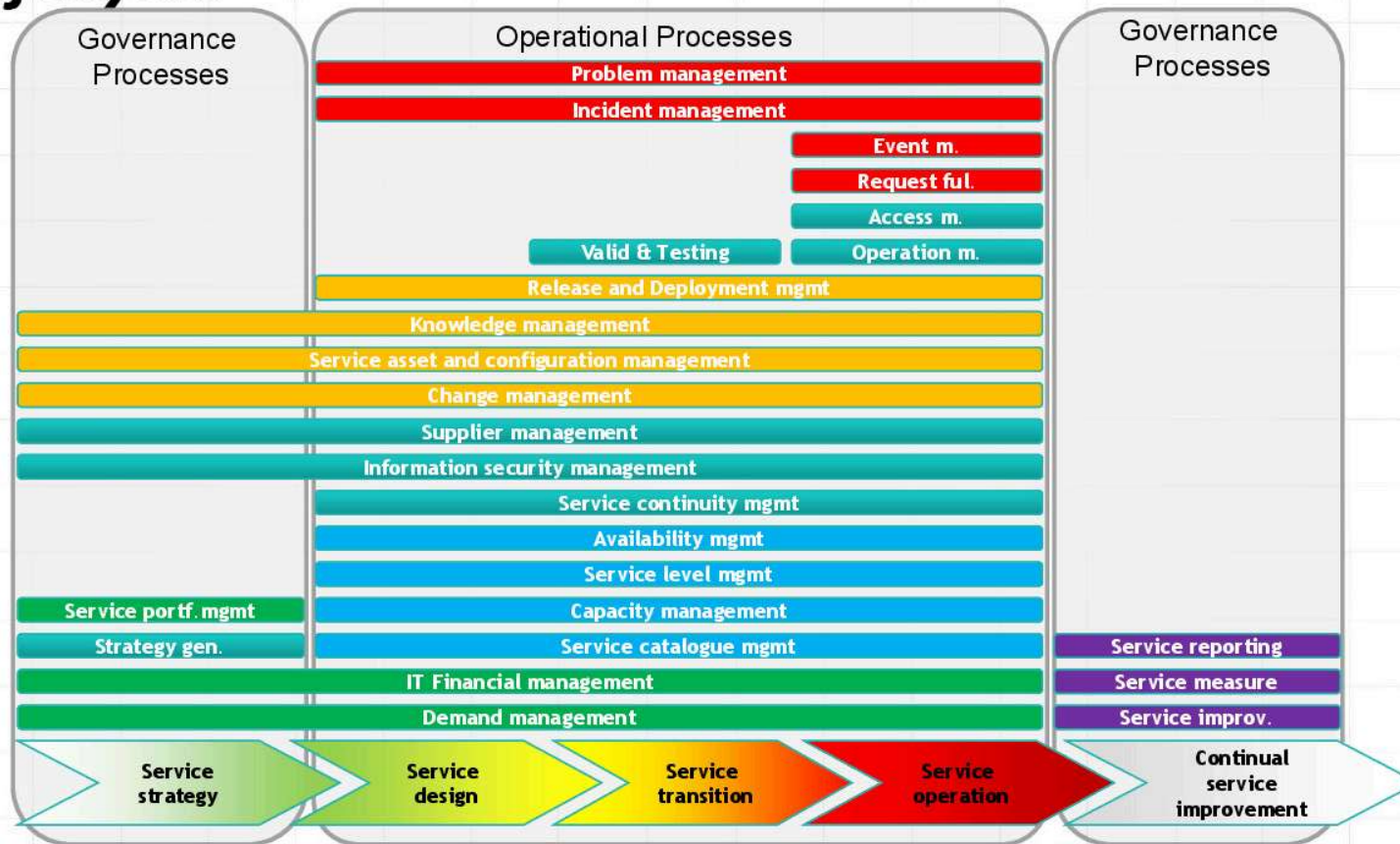
- Design Coordination
- Service Catalog Management
- **Service Level Management**
- Availability Management
- Capacity Management
- IT Service Continuity Management
- Information Security Management
- Supplier Management

SERVICE TRANSITION

- Transition Planning and Support
- **Change Management**
- **Service Asset & Configuration Management**
- **Release & Deployment Management**
- Service Validation
- Change Evaluation
- **Knowledge Management**



Service Lifecycle



Incident Management

- **Incident** – unplanned interruption to an IT service or an unplanned reduction in its quality
- Process aim: remove the incident as soon as possible to restore the agreed service level.
 - Incident diagnosis
 - registration
 - categorization
 - prioritization



Problem Management



- **Problem** – underlying cause of one or more incidents
- Process aim: to find & remove the root cause of one or more incidents through a change (**Request For Change**).
- A **workaround** = temporary solution.
- Symptoms of incidents and their workarounds are **known errors**.



Service Asset & Configuration Management

- Process aim: to build and maintain (**Configuration Management DataBase -> CMDB**) with resources (**CIs**) needed for services delivery
- CMDB database stores the description of the configuration item and their mutual relations at the level of detail appropriate to the needs



Change Management

- **Configuration Item (CI)** - a single element needed for service delivery
- **Request For Change** of one or more configuration elements (CI), which is the basis for starting the change management process
- Process aim: to ensure that changes are implemented in a controlled manner with minimal disruption to business



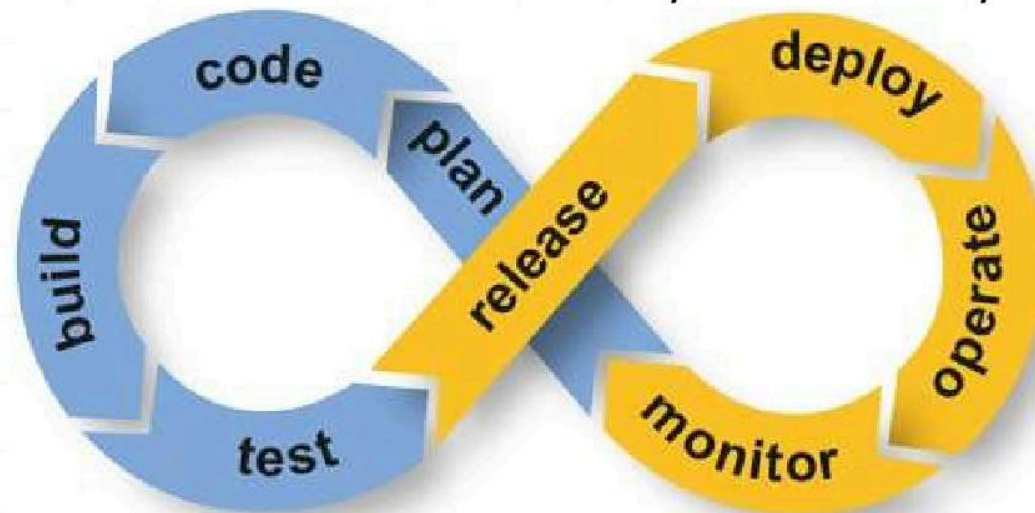
Knowledge Management

- Process aim: right decision making, knowledge transfer, qualification management, etc.
- As part of this process, **Knowledge Management System** is maintained which should be integrated with the CMDB database

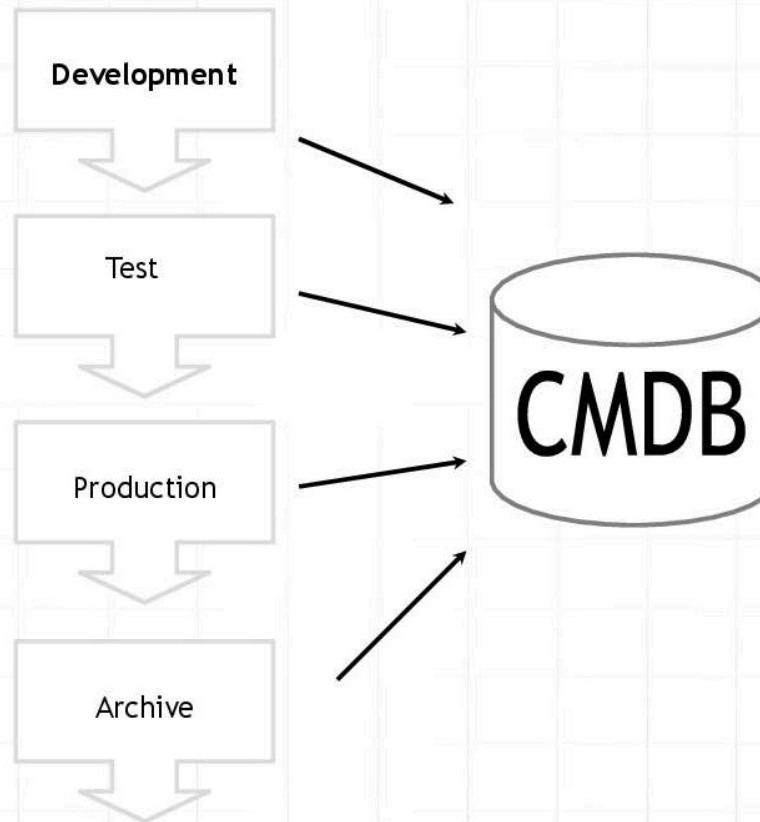


Release & Deployment

- Process aim: technical build of software package, tests and implementation of a service or a change
- ensure that the recipient will use the service efficiently and efficiently



Release & Deployment



Event Management

- **Event** – any registerable status change of a CI needed to perform the service
- Process aim:
 - monitoring
 - registration
 - categorization of events in IT environment that provides services



Service Level Management

- Process aim: an agreement to ensure the terms of providing a specific service to a specific recipient
- These conditions are recorded in the service contract - ***Service Level Agreement***
- The starting points
 - standard conditions specified in ***Service Catalog***
 - ***Service Level Requirements*** of a service recipient

Metrics & KPIs

- You can't **control** what is **undefined**
- You can't **measure**, what is **not under control**
- You can't **improve**, what is not **measured**



Metrics & KPIs



- **Key Performance Indicators** - measurable values indicating how effectively a company achieves key business goals
- **KPIs** provide the most important information about **performance**, allowing to understand whether the organization is on track to achieve its goals (or not)

Metrics & KPIs

- Example

The business goal: to manufacture annually 20 000 SSDs (**KPI**)

Current status: 5 000 have been produced since for 10 months (**Metric**)

$$5\ 000 / 20\ 000 * 100\% = 25\ \%$$

Conclusion: It is unlikely that the goal will be achieved

Continual Service Improvement

- Activities:
 - monitoring compliance with processes and procedures
 - monitoring the effectiveness of services and processes
 - monitoring the performance of services and processes
- **Deming cycle** (*Plan-Do-Check-Act cycle*)
 - A diagram illustrating the basic principle of continuous improvement
 - proposed by William E. Deming



Continual Service Improvement

- **Plan** – plan who, what, when and how to do it
- **Do** – make planned changes
- **Check** – check the results
- **Act** – update the plan according to the results



4 P's of Service Management

People

- skills, training, communication

Processes

- actions, activities, changes, goals, improving paths

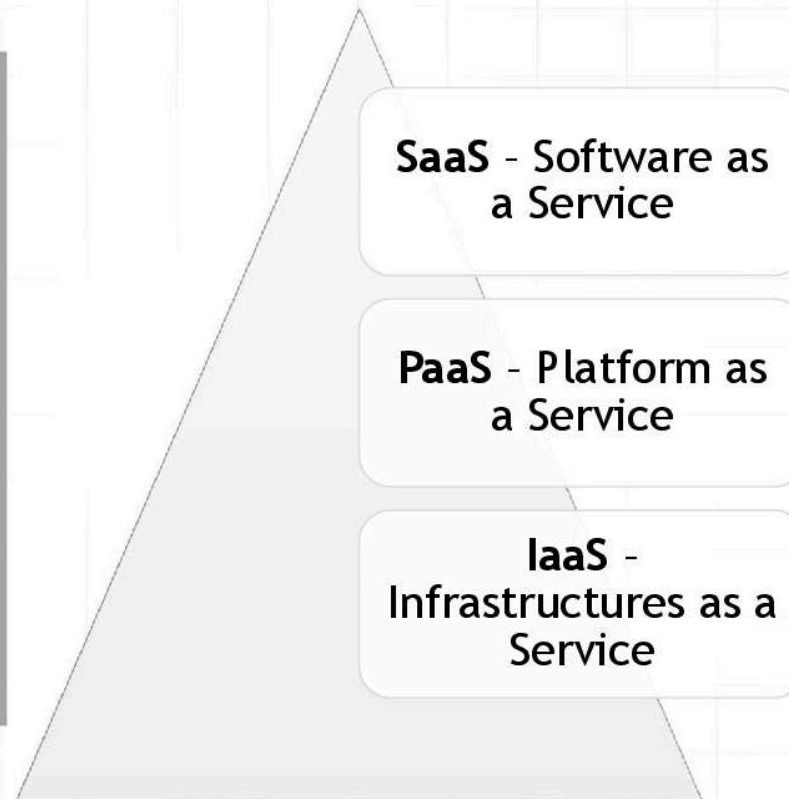
Products

- tools, monitors, measures, documents

Partners

- vendors, specialists, suppliers

Service Delivery Levels



Service Delivery Levels

■ User managed ■ Provider managed

On premises	IaaS	PaaS	SaaS
Application	Application	Application	Application
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
Operating system	Operating system	Operating system	Operating system
Virtualization	Virtualization	Virtualization	Virtualization
Networking	Networking	Networking	Networking
Storage	Storage	Storage	Storage
Servers	Servers	Servers	Servers

Why ITIL makes sense?

- introduces a common language to understand pages (IT and business)
- enables the service provider to be adapted to the needs of the service recipient
- allows to agree realistic services levels that deliver necessary value at an acceptable cost

Why ITIL makes sense?

- defines specific processes with documented responsibility for each activity
- processes can be monitored based on Service Level Agreements (SLAs) and improved if necessary (CSI)